AccessiFlow Platform KT & Architecture Walkthrough

This document outlines the architecture, tooling, and development pipeline for the **AccessiFlow** digital inclusion platform as developed by Prasan Kumar.

---

1. Executive Summary

AccessiFlow is a futuristic, highly interactive React-based Single Page Application. It is designed to act as a simulated AI-auditing engine for WCAG 2.1 & 2.2 AA validation. The application simulates complex auditing capabilities—such as DOM traversal, Contrast Ratio checking, and ARIA tree building—and can export consolidated testing results into "Master Audit" Excel `.xlsx` spreadsheets natively within the browser.

---

2. Technical Stack and Architecture

The platform relies on a modern build toolchain oriented towards speed, small bundles, and visually striking UX design.

- **Frontend Core:** React 18
- **Build Engine & Compilation:** Vite 8.0.3 + Node 22 (for strict ECMA script builds)
- **Styling Architecture:** TailwindCSS v4 with "Zero-Gravity" themed design tokens (Deep Space `#0d0d0d`, Brand Blue `#0076d6`, Accent Red `#ff4f52`).
- **Animations:** `framer-motion` for complex keyframes, modal mounting (`AnimatePresence`), and SVG loader ring strokes.
- **Routing:** `react-router-dom` mapping `/` to the `LandingPage` and `/dashboard` to the `Dashboard` scanner view.
- **Asset Generation:** `xlsx` library parsing 2D arrays directly into Office-spec spreadsheets dynamically for the Master Audit.

---

3. Visual Walkthrough & Core Components

3.1 The Landing Experience

The root component (`/`) hosts the `LandingPage.jsx`. It includes a live-simulated issue counter ticking up and a feature matrix outlining the "Core 4" capabilities.

Figure 1: AccessiFlow Landing Page

A11y Issue Detection Agent: 151 potential issues blocked

Achieve 360° Digital Inclusivity with AI-Powered Accessibility Testing

From automated web scans to complex PDF remediation and manual screen-reader audits—ensure your digital products meet WCAG 2.1/2.2 AA standards with the industry's most comprehensive toolkit.

Start Free Scan

Book a Demo →

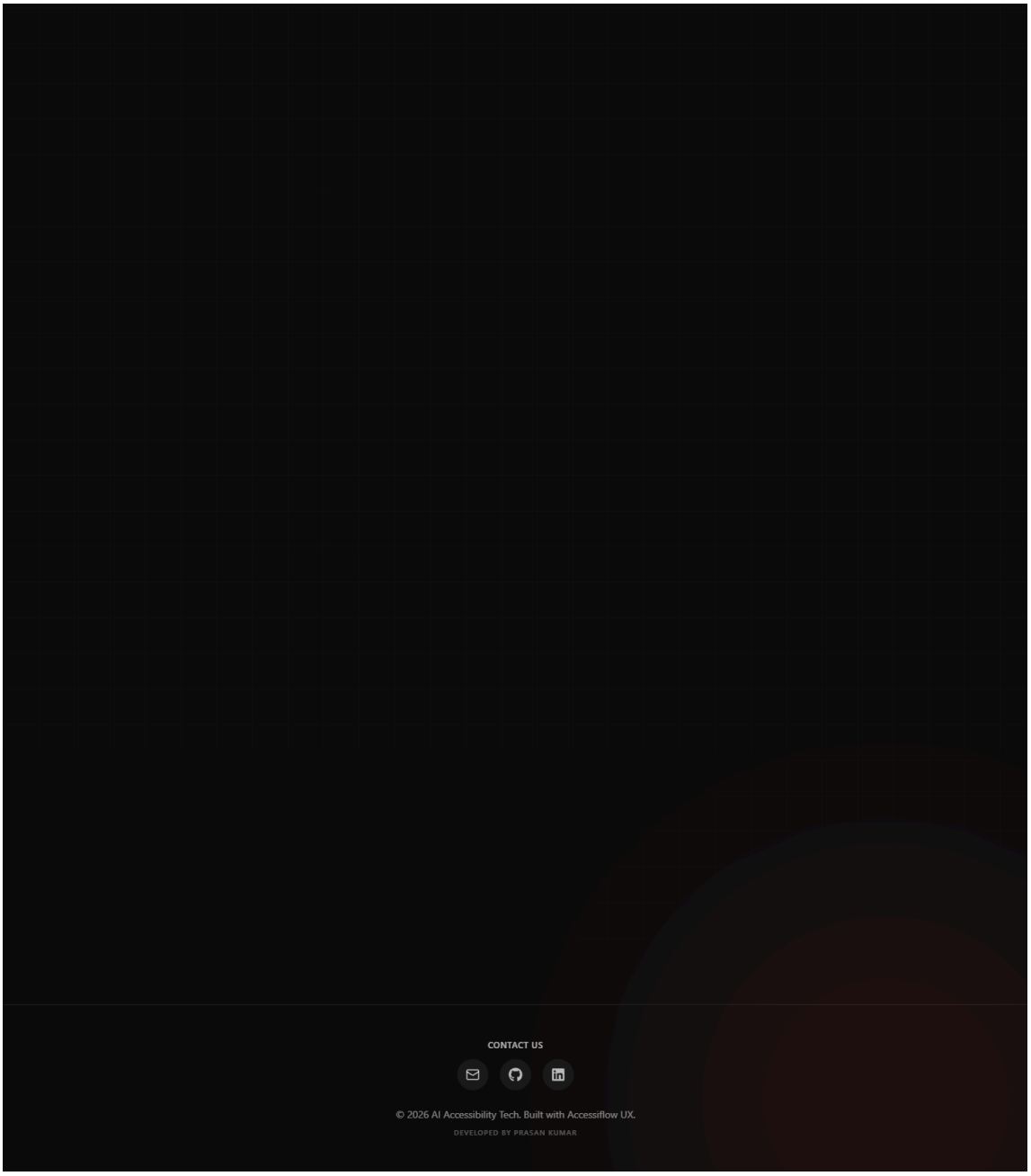
WCAG 2.1 & 2.2 AA Compliant

The Core 4 Toolkit

Everything you need, unified in one platform.

The Technical Tooling

Deep integration with industry-standard accessibility tools.

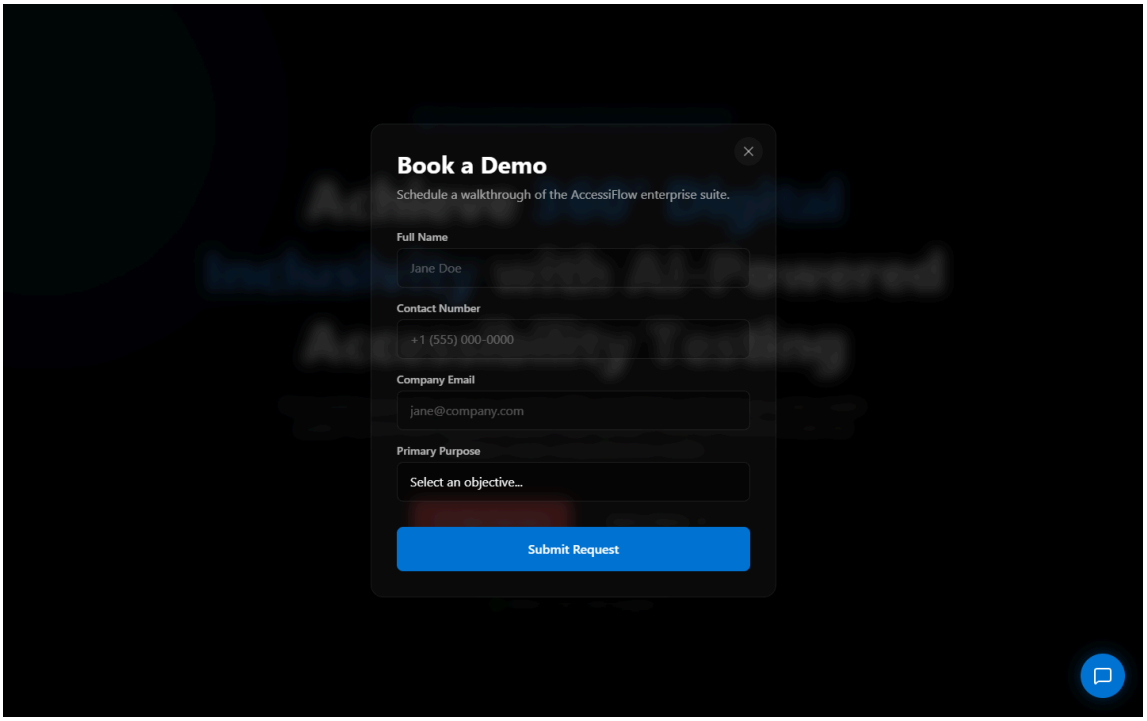


At the bottom of this page, contact logic maps directly to GitHub, LinkedIn, and Email anchors embedded neatly inside the footer components.

3.2 Lead Generation (The Modal System)

Attached to the Hero component is the **Book a Demo** trigger. Clicking this engages a deep `AnimatePresence` Framer window that prevents users from interacting with the background.

Figure 2: Form Interaction Modal

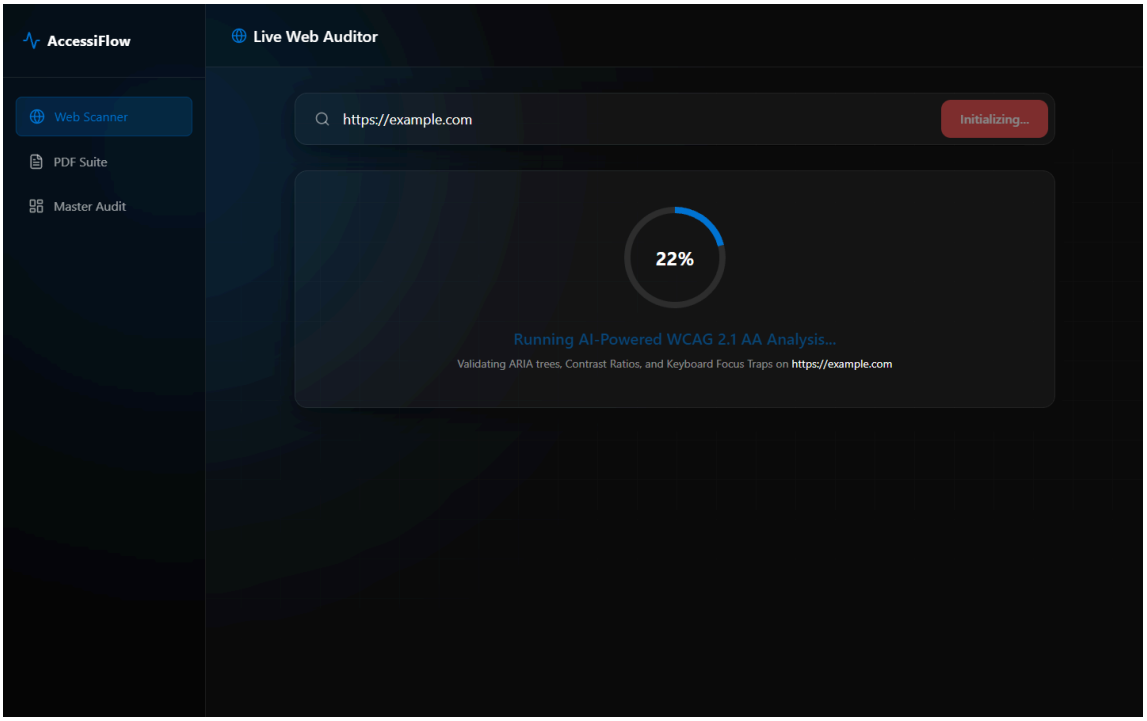


Key Logic Highlight: We implemented direct internal `useState` validation filtering out generic email patterns (like `@gmail.com` or `@icloud.com`) ensuring only company domains can register intent.

3.3 The Core Dashboard Simulator

If a user hits "Start Free Scan", `useNavigate()` bumps them to the `/dashboard` route.

Figure 3: Dashboard Scanner Interface



The scanning logic is a complex illusion. When a URL is submitted:

1. React sets `scanning=true`, locking inputs.
2. A `setInterval` block randomly increments a percentage counter while manipulating the `strokeDashoffset` property of a 377-point SVG circle.
3. Upon hitting 100%, the component unmounts the scanner loader and immediately mounts random pre-seeded "Critical" and "High" WCAG constraint violations arrays.

---

4. Master Audit Excel Generation

Inside the simulated payload output or specifically inside the `ReportingFeature.jsx`, users can download the mock **Master Audit**.

The code uses the `xlsx` utility function `aoa_to_sheet`:

```
const handleDownloadSample = () => {
    const sheetData = [
      ["Audit Date", new Date().toLocaleString()],
      ["Target URL", "https://accessiflow.org"],
      ...
    ];
    const worksheet = XLSX.utils.aoa_to_sheet(sheetData);
    const workbook = XLSX.utils.book_new();
    XLSX.utils.book_append_sheet(workbook, worksheet, "Sample Audit");
```

```
    XLSX.writeFile(workbook, `accessiflow-sample-report.xlsx`);
};
```

This bypasses backend logic completely, dumping a natively written file to the OS.

---

5. Google Cloud Run Deployment Protocol

The repository is built strictly to integrate via Google Cloud Container Registry.

Dockerfile Architecture:

```
# Stage 1: Build the Vite React App
FROM node:22-alpine AS builder  # Strict compatibility with Vite 8
WORKDIR /app
COPY package.json package-lock.json ./
RUN npm ci
COPY . .
RUN npm run build

# Stage 2: Serve the app using NGINX
FROM nginx:alpine
COPY nginx.conf /etc/nginx/conf.d/default.conf # Maps ALL / routes to index.html for React
Router
COPY --from=builder /app/dist /usr/share/nginx/html
EXPOSE 8080
```

Deployment Commands:

```
gcloud run deploy accessiflow-landing \
   --source . \
   --project linkedin-landing-page-491707 \
   --region us-central1 \
   --allow-unauthenticated
```

It currently routes directly to `https://accessiflow-landing-643526693397.us-central1.run.app`.

Validation Sign-off

This completes the visual walkthrough and architecture layout. The application is completely version-controlled, functional, containerized, and publicly accessible.

Submitted by Prasan Kumar